

Chapter 6: Line graphs and Edge-coloring

6.1 Line Graphs

The **line graph** $L(G)$ of G has as vertex set the edges of G , with two vertices of $L(G)$ adjacent if the corresponding edges of G share an end-vertex. For example, a matching in G corresponds to an independent set in $L(G)$. It is easy to show that a line graph is **claw-free**, meaning it does not have an induced copy of $K_{1,3}$.

For each vertex v of the original graph G , the edges of G incident with v become a clique in $L(G)$. Further, each edge in G has two ends, and so is part of two cliques in $L(G)$. It turns out that the converse is true:

KRAUSZ DECOMPOSITION. *A graph H is the line graph of some simple graph if and only if one can decompose the edges of H into cliques such that every vertex is in at most two cliques.*

There are graphs that are claw-free but not line graphs, for example the wheel W_5 . In fact, Beineke showed that there are eight graphs that are minimal with respect to the property of being claw-free but not line graphs. This implies a polynomial-time algorithm for determining whether a graph is a line graph. (More on algorithms later.)

6.2 Edge-Chromatic Number

The edge-chromatic number (or chromatic index) χ' of a graph is the minimum number of colors needed to color the edges such that no two edges that share a vertex have the same color. This equals $\chi(L(G))$. It is immediate that $\chi'(G) \geq \Delta(G)$.

THE COMPLETE GRAPH. $\chi'(K_n)$ is $n - 1$ if n is even and n otherwise.

PROOF. Clearly $n - 1$ is a lower bound. It can be observed that an r -regular graph has $\chi' = r$ if and only if the edges can be decomposed into r perfect matchings. Therefore $n - 1$ is not possible when n is odd. The upper bound is left as an exercise. $\square$

König showed that if a graph is bipartite then $\chi'(G) = \Delta(G)$. (We already observed this for regular bipartite graphs.) A deeper result is:

VIZING/GUPTA. *For any simple graph G it holds that $\chi'(G) \leq \Delta + 1$.*

6.3 The List Chromatic Number

Suppose each vertex v of a graph has a list/palette L_v of colors. A valid coloring of a graph means give each vertex a color from its list such that the result is a proper coloring. The **list chromatic number** is then the smallest integer k such that for every possible collection of lists of size k , it is possible to find a valid coloring. Denoted χ_ℓ . The ordinary chromatic number corresponds to the case where the lists are all the same. Thus $\chi_\ell(G) \geq \chi(G)$.

For example, $\chi_\ell(C_5) = 3$. We say a graph is **k -choosable** if the list chromatic number is at most k . For example, the greedy algorithm shows that G is $(\Delta + 1)$ -choosable.

COMPLETE BIPARTITE. For example, $K_{3,3}$ is not 2-choosable. For each side, make one vertex have list $\{1, 2\}$, one list $\{1, 3\}$ and one $\{2, 3\}$. Then to color the vertices on one side one has to use at least two colors, which means four colors in total (since the colors on the two sides have to be disjoint), a contradiction. Also $K_{2,4}$ is not 2-choosable. It can be shown that $K_{7,7}$ is not 3-choosable: for each side, give each vertex one of the blocks/lines of the Fano plane (the decomposition of K_7 into K_3 's). The exact value of $\chi(K_{n,n})$ is in general not known, but is asymptotically $\log_2 n$.

The most major open problem is the edge-coloring version:

CONJECTURE. *For every graph, $\chi'_\ell(G) = \chi'(G)$.*

Galvin proved equality when the graph is bipartite.